VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Microcontroller

Lab Report

Lab 1: LED Animations

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1 Exercise 1

Report 1: Schematic

The Proteus schematic for this exercise is shown below.

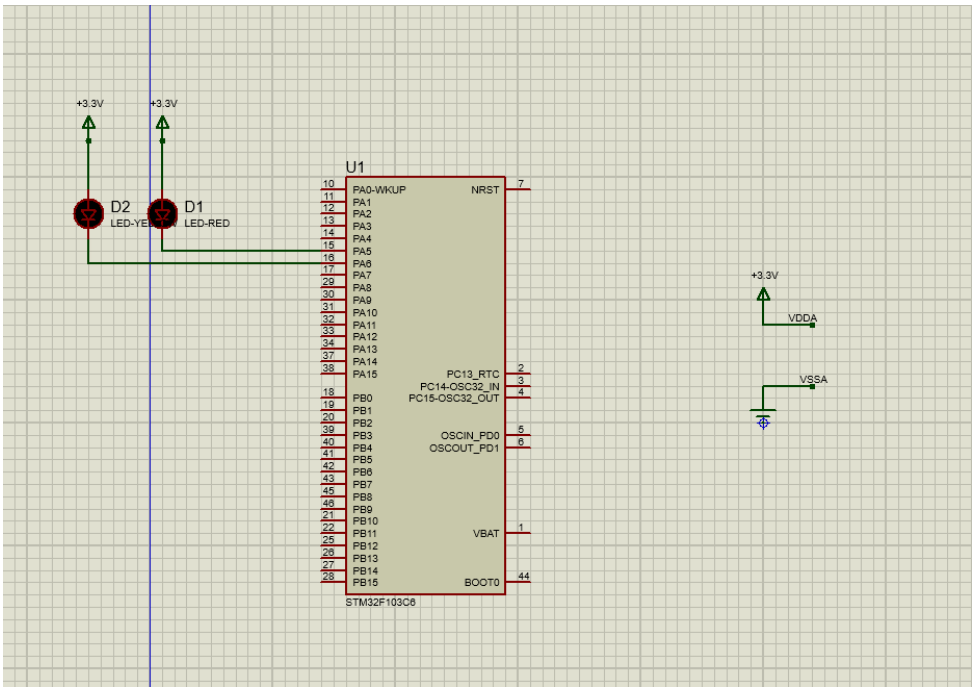


Figure 1.1: https://github.com/aulequangtri/microprocessor_lab

Report 2: Source Code

The source code inside the `while(1)` loop to switch the states of the RED and YELLOW LEDs every 2 seconds:

```
<MINTED>
```

2 Exercise 2

Report 1: Schematic

The Proteus schematic for the 3-LED traffic light system is shown below.

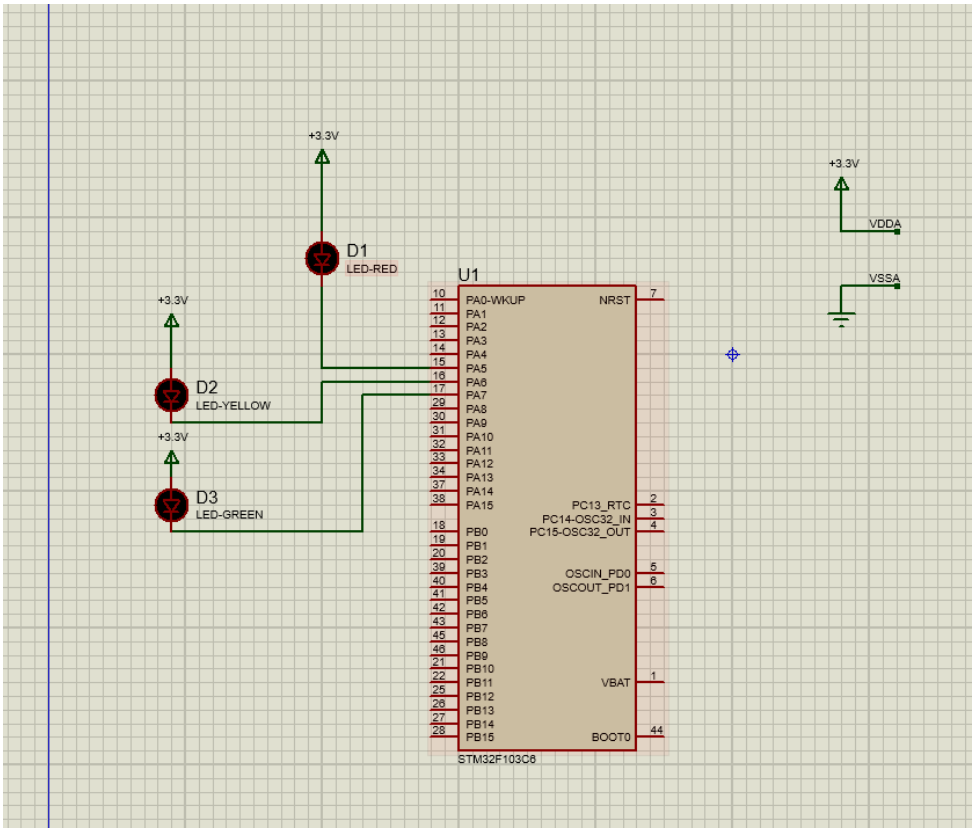


Figure 2.1: Schematic for Exercise 2 (Traffic Light)

Report 2: Source Code

The source code inside the `while(1)` loop simulating the traffic light cycle:

```
<MINTED>
```

3 Exercise 4

Report 1: Schematic

The schematic for the 7-segment display integration.

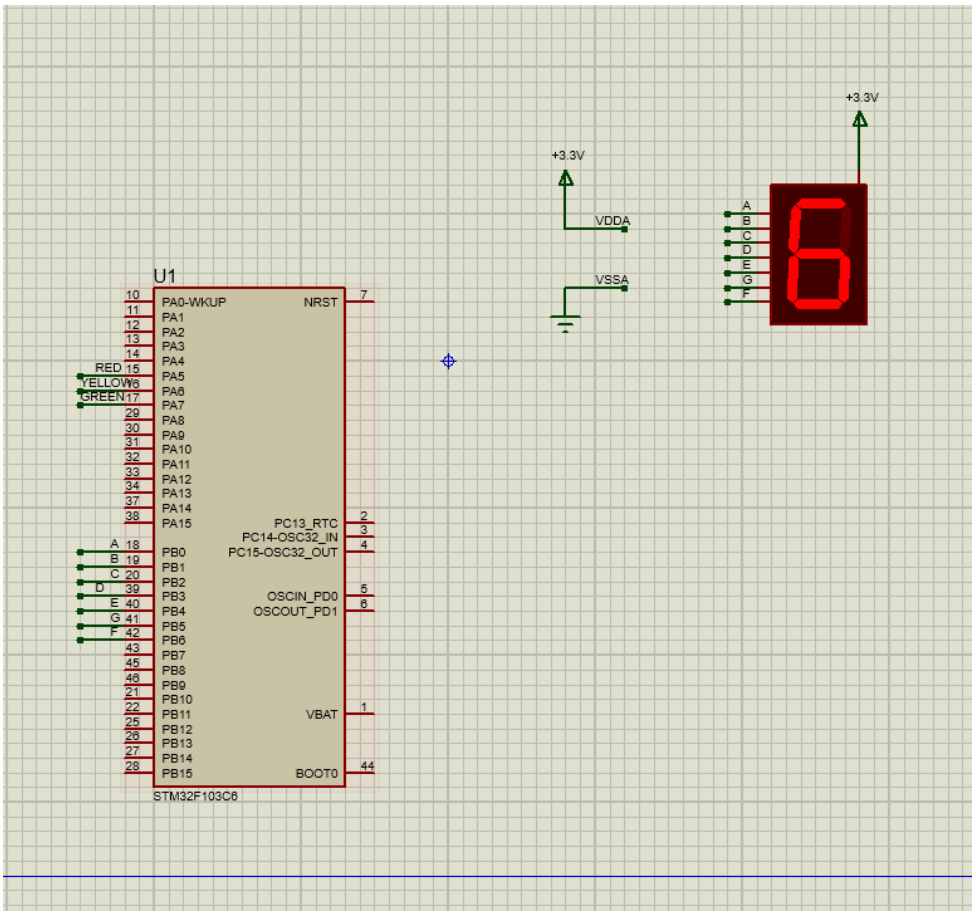


Figure 3.1: Schematic for Exercise 4 (7-Segment LED)

Report 2: Source Code for display7SEG function

The function `display7SEG(int num)` is implemented as follows:

```
<MINTED>
```

4 Exercise 5

Report 1: Schematic

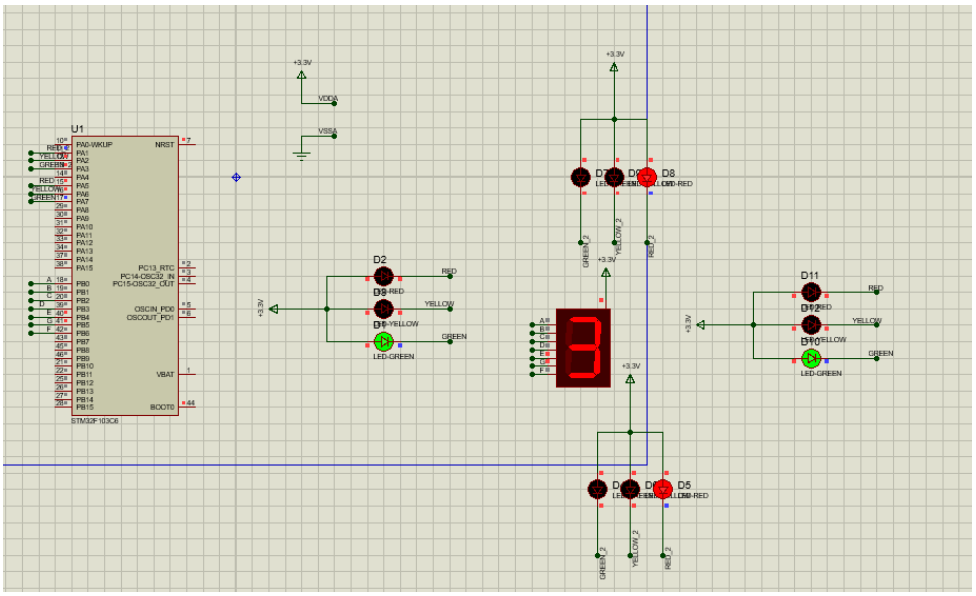


Figure 4.1: Schematic for Exercise 5 (4-Way Traffic Light + 7SEG)

Report 2: Source Code

The code snippet inside the infinite loop for the 4-way traffic light integrated with a 7-segment countdown:

```
<MINTED>
```

5 Exercise 6

Report 1: Schematic

The Proteus schematic for the analog clock using 12 LEDs.

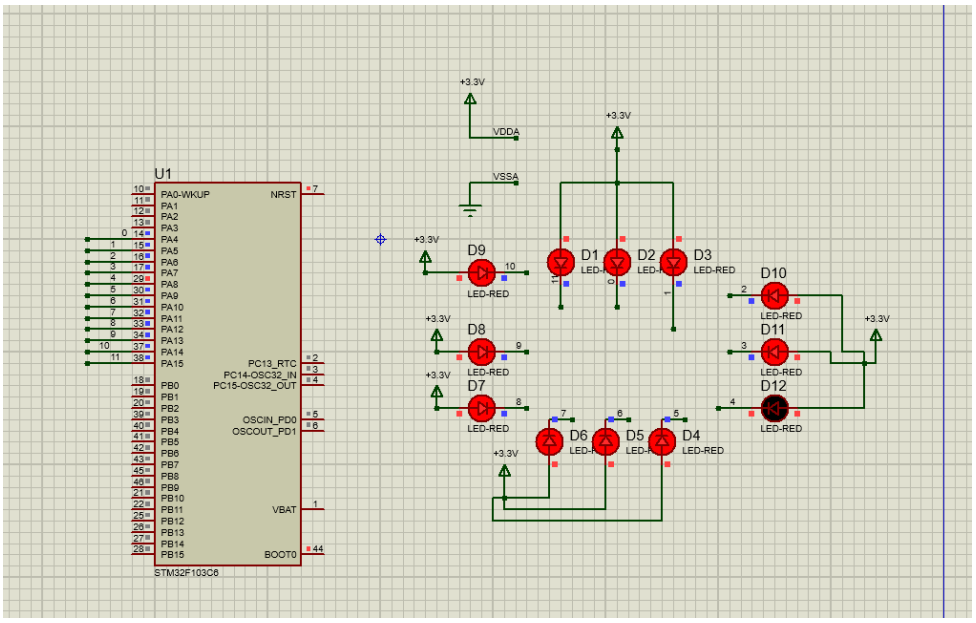


Figure 5.1: Schematic for Exercise 6 (Analog Clock)

Report 2: Source Code

The testing program turning every LED in a sequence:

```
<MINTED>
```

6 Exercise 7 to 10

The following schematic is used for the analog clock exercises (7 to 10):

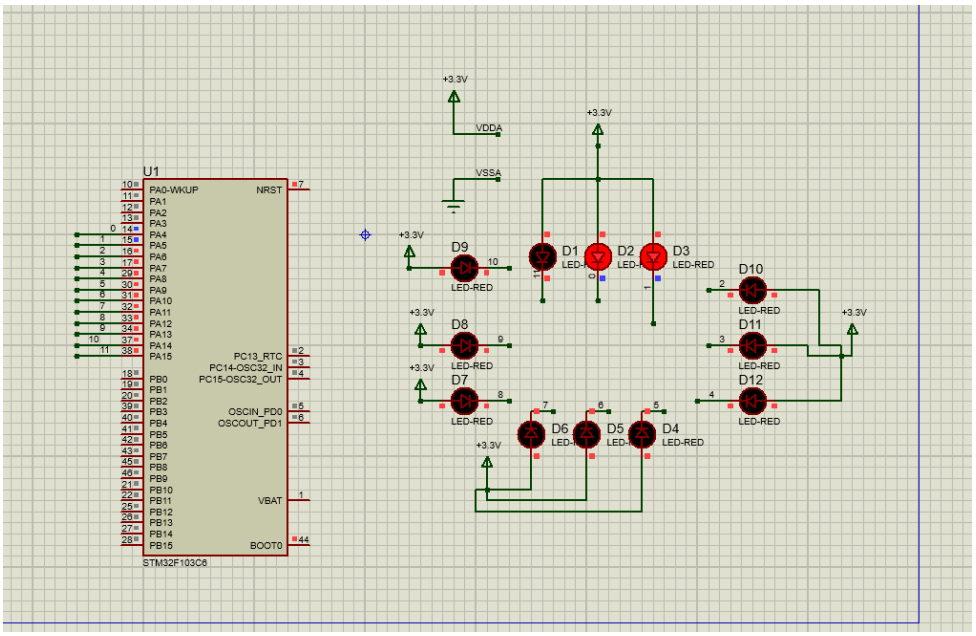


Figure 6.1: Schematic for Exercises 7-10 (Analog Clock Functions)

Exercise 7: clearAllClock()

```
<MINTED>
```

Exercise 8: setNumberOnClock(int num)

```
<MINTED>
```

Exercise 9: clearNumberOnClock(int num)

```
<MINTED>
```

Exercise 10: Analog Clock Main Loop

The integration of the system using 12 LEDs to display an analog clock (Hour, Minute, and Second).

```
<MINTED>
```